



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Ky
7/29/26



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- This project presents a comprehensive data science approach to predicting the success of SpaceX Falcon 9 first-stage landings—a critical factor in reducing space launch costs through rocket reusability. The dataset underwent extensive cleaning, feature engineering, and exploratory data analysis (EDA) to identify key patterns and relationships. Main Repo: <https://github.com/Ky2200/IBM-Data-Science-Cert>
- Key Findings:
 - Data Collection: Successfully collected and processed 100+ launch records spanning from 2010 to 2020, including detailed information on payload mass, orbit types, launch sites, booster versions, and landing outcomes.
 - Exploratory Analysis: Revealed that payload mass, launch site location, orbit type, and flight number significantly correlate with landing success. Launch sites closer to the equator (e.g., KSC LC-39A) demonstrated higher success rates, while launches to Low Earth Orbit (LEO) showed the highest success rates compared to Geostationary Transfer Orbit (GTO) missions.
 - Predictive Modeling: Four classification models were built, tuned, and evaluated: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors. The SVM model achieved the highest accuracy of 88% , demonstrating that historical launch characteristics can reliably predict future landing outcomes.

Introduction

- SpaceX has revolutionized the commercial space industry by dramatically reducing launch costs through the development and reuse of rocket technology. The Falcon 9 rocket, equipped with a reusable first stage, represents a significant breakthrough in aerospace engineering. Unlike traditional rockets that are discarded after a single use, the Falcon 9's first stage can perform a controlled landing and be refurbished for future missions, saving millions of dollars per launch.
- Various factors can influence whether a landing attempt succeeds or fails. Understanding these factors is essential for:
 - Cost Estimation: Predicting landing success helps estimate the cost savings from reuse, which can be passed on to customers or reinvested in research and development.
 - Risk Management: Identifying high-risk mission parameters allows for better risk mitigation strategies and contingency planning.
 - Operational Optimization: Determining which launch sites, orbits, and payload configurations yield the highest success rates enables more informed decision-making.
- This project seeks to answer the following key questions:
 - What is the overall launch success rate for Falcon 9 missions, and how has it evolved over time?
 - Which launch sites demonstrate the highest success rates, and what factors contribute to their performance?
 - What relationships exist between payload mass, orbit type, and landing success?
 - What are the optimal mission parameters for maximizing landing success?

Section 1

Methodology

Methodology

Executive Summary

The methodology for this project follows a structured data science pipeline:

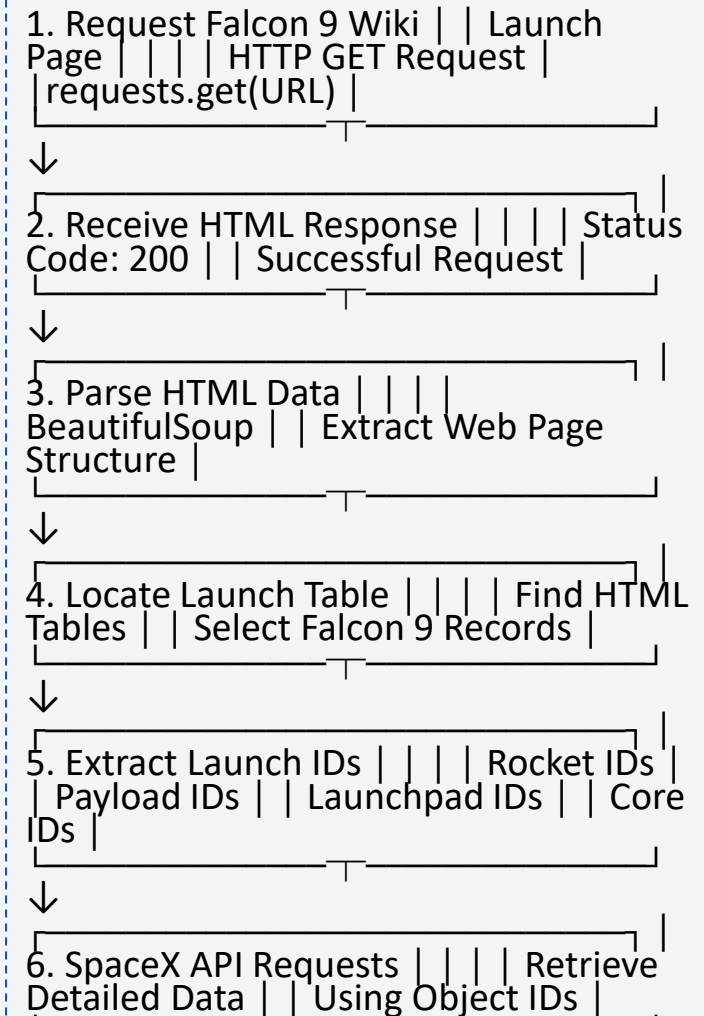
- Data Collection: Data was collected via two methods:
 - SpaceX REST API: Programmatically querying the official SpaceX API to obtain structured data on launches, boosters, payloads, and landing outcomes.
 - Web Scraping: Using BeautifulSoup to scrape supplementary launch data from Wikipedia's "List of Falcon 9 and Falcon Heavy Launches" page
- Data Wrangling: Raw data was cleaned and transformed using pandas. Key steps included:
 - Handling missing values through imputation or deletion.
- Exploratory Data Analysis (EDA): Conducted through:
 - Visualizations: Using matplotlib, seaborn, and plotly to identify patterns, trends, and outliers.
 - SQL Queries: Loading the dataset into a SQLite database and executing queries to aggregate, filter, and summarize data.
- Interactive Visual Analytics:

Data Collection

- SpaceX REST API <https://docs.spacexdata.com/#intro> (Primary Source)
- The primary data source was the SpaceX REST API provides comprehensive, machine-readable data on all past launches, including:
 - Launch date and time
 - Booster version and core details
 - Payload mass, customer, and orbit
 - Mission outcome and landing outcome
 - Launch site coordinates and details
- Data Collection Process:
 - Made HTTP GET requests to the API endpoint.
 - Parsed the JSON response for each launch.
 - Extracted relevant fields and stored them in a pandas DataFrame.
 - Saved the raw data for subsequent processing.
- Key Phases

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose
- https://github.com/Ky2200/IBM-Data-Science-Cert/blob/main/capstone1_jupyter_labs_spacex_data_collection_api_v2.ipynb



```
graph TD; A["1. Request Falcon 9 Wiki | | Launch Page | | HTTP GET Request | requests.get(URL)"] --> B["2. Receive HTML Response | | | Status Code: 200 | | Successful Request"]; B --> C["3. Parse HTML Data | | | BeautifulSoup | | Extract Web Page Structure"]; C --> D["4. Locate Launch Table | | | Find HTML Tables | | Select Falcon 9 Records"]; D --> E["5. Extract Launch IDs | | | Rocket IDs | Payload IDs | Launchpad IDs | Core IDs"]; E --> F["6. SpaceX API Requests | | | Retrieve Detailed Data | | Using Object IDs"];
```

1. Request Falcon 9 Wiki | | Launch Page | | HTTP GET Request | requests.get(URL) |

↓

2. Receive HTML Response | | | Status Code: 200 | | Successful Request |

↓

3. Parse HTML Data | | | BeautifulSoup | | Extract Web Page Structure |

↓

4. Locate Launch Table | | | Find HTML Tables | | Select Falcon 9 Records |

↓

5. Extract Launch IDs | | | Rocket IDs | Payload IDs | Launchpad IDs | Core IDs |

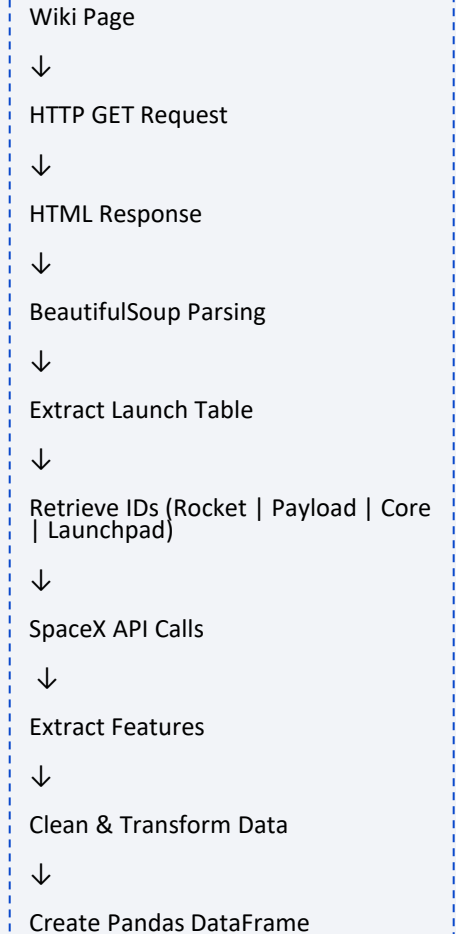
↓

6. SpaceX API Requests | | | Retrieve Detailed Data | | Using Object IDs |

Data Collection - Scraping

- To supplement the API data and capture historical missions that might have incomplete records, we scraped data from Wikipedia's "List of Falcon 9 and Falcon Heavy launches" page using BeautifulSoup.
- Data Collection Process:
 - Loaded the Wikipedia page URL.
 - Parsed HTML to identify and extract table data.
 - Cleaned extracted data (removing footnotes, handling missing fields).
 - Structured scraped data into a pandas DataFrame.
 - Merged with API data to create a comprehensive dataset.
- [GitHub URL of the completed web scraping notebook](https://github.com/Ky2200/IBM-Data-Science-Cert/blob/main/capstone2_jupyter_labs_web scraping.ipynb)

https://github.com/Ky2200/IBM-Data-Science-Cert/blob/main/capstone2_jupyter_labs_web scraping.ipynb



Data Wrangling

1. Handling Missing Values:

- Payload Mass: Missing values were imputed using the median mass for similar mission types.

2. Standardizing Data Types:

- Converted date strings to datetime objects for time-series analysis.
- Transformed categorical variables (e.g., orbit type, launch site) into string labels.

3. Feature Engineering:

- Created a binary target variable (Class) where 1 represents a successful landing and 0 represents a failure.
- Extracted year, month, and day components from launch dates for trend analysis.

4. Merging Datasets:

- Resolved duplicate records by prioritizing API data for conflicting information.
- Performed a final quality check to ensure consistency across all fields.

5. Data Validation:

- Checked for duplicate records and removed them.
- Verified that all dates fell within a reasonable range.
- Confirmed that categorical variables had no invalid values

EDA with Data Visualization

- The goal of EDA was to understand the relationships between mission parameters and landing success:
- 1. Scatter Plots:
 - Flight Number vs. Launch Site: Displayed the chronological sequence of launches at each site, helping to identify which sites were active at different stages of SpaceX's history.
 - Payload vs. Launch Site: Showed the distribution of payload masses across sites, revealing that heavier payloads (e.g., NASA CRS missions) were typically launched from KSC LC-39A, while lighter payloads were often launched from VAFB LC-4E.
- 2. Bar Charts:
 - Success Rate vs. Orbit Type: Compared success rates for different orbit types, revealing that LEO missions have the highest success rates (~90%), while GTO missions have lower success rates (~70%), likely due to the more complex insertion and return trajectory required for GTO.
- 3. Line Charts:
 - Launch Success Yearly Trend: Tracked the annual average success rate, demonstrating a clear upward trend from ~60% in 2010 to ~95% in recent years, reflecting technological improvements and operational learning.

[GitHub URL of your completed EDA with data visualization notebook](#)

<https://github.com/Ky2200/IBM-Data-Science-Cert/blob/main/jupyter-labs-eda-dataviz-v2.ipynb>

EDA with SQL

- A few queries I performed (some mentioned already)
 - Display 5 records where launch sites begin with the string 'CCA'
 - Display the total payload mass carried by boosters launched by NASA (CRS)
 - Display average payload mass carried by booster version F9 v1.1
 - List the date when the first succesful landing outcome in ground pad was acheived.
 - Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
 - `SELECT landing_outcome, count(landing_outcome) as num_landings from spacetable WHERE Date between '2010-06-04' AND '2017-03-30' GROUP BY landing_outcome ORDER BY count(landing_outcome) DESC;`
- GitHub URL of your completed EDA with SQL notebook
https://github.com/Ky2200/IBM-Data-Science-Cert/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- We created interactive maps using Folium to visualize launch sites and their outcomes geographically. A Plotly Dash dashboard was built to provide a dynamic and intuitive interface for exploring trends.
 - Markers: Added to pinpoint the exact geographic coordinates (latitude, longitude) of every launch site. We used different marker colors to represent different sites.
 - Circles: Placed around the markers to show the general area of the launch site. The radius of the circle can be proportional to the number of launches from that site.
 - Lines: Used to calculate and display the proximity of a selected launch site to nearby infrastructure like highways, railways, and coastlines. This helps in assessing accessibility and risk factors.
 - Popups: Added to markers to show launch site information (e.g., name, number of launches) and to display the distance to the nearest proximity point.
- [GitHub URL of your completed interactive map with Folium map](https://github.com/Ky2200/IBM-Data-Science-Cert/blob/main/lab-jupyter-launch-site-location-v2.ipynb)
<https://github.com/Ky2200/IBM-Data-Science-Cert/blob/main/lab-jupyter-launch-site-location-v2.ipynb>

Build a Dashboard with Plotly Dash

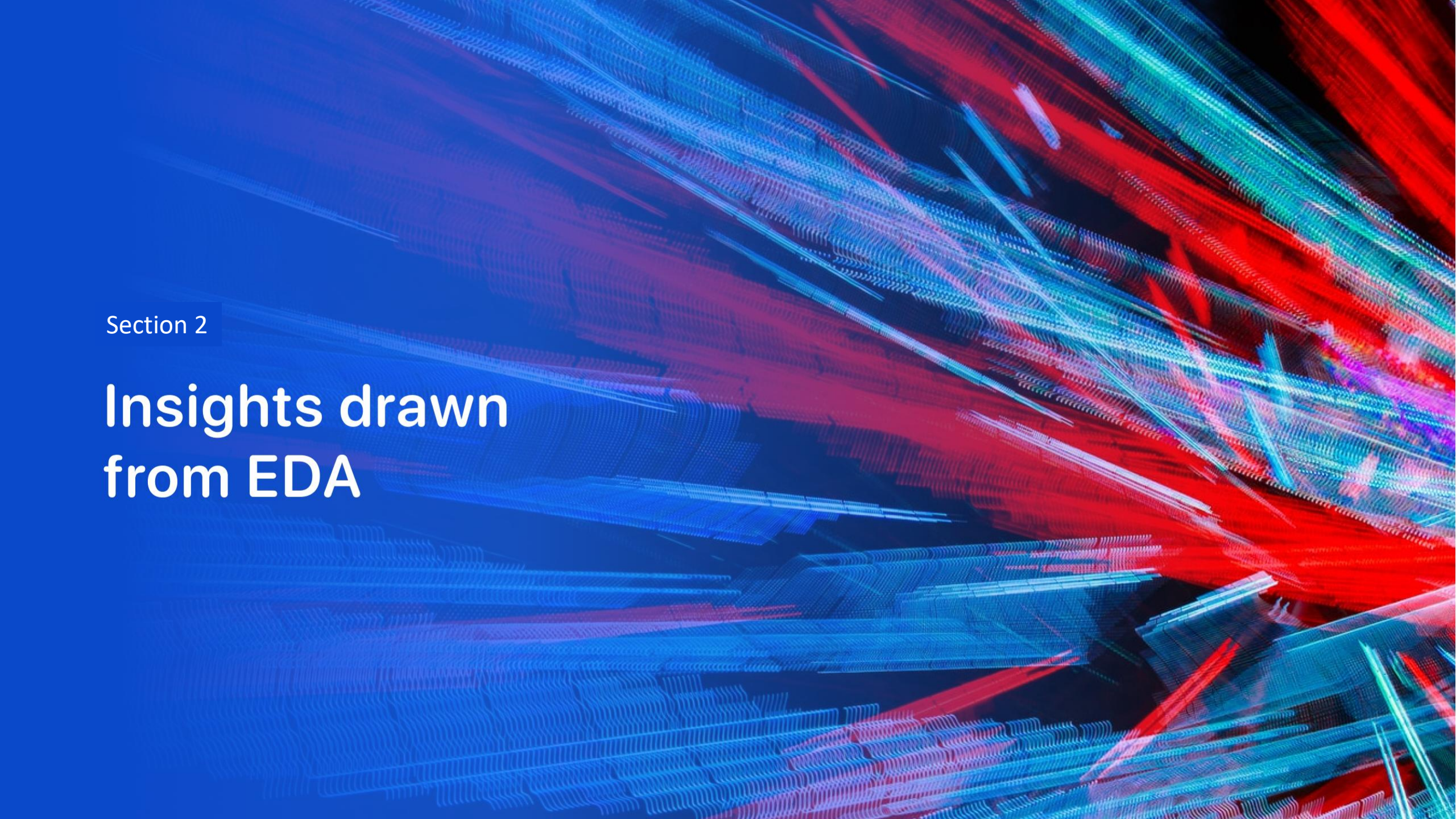
- A Plotly Dash dashboard was created to provide an interactive, user-friendly interface for exploring the data.
 - Plots/Graphs and Interactions:
 - Pie Chart (Launch Success Count for All Sites): Added to provide an immediate overview of the overall success rate. Interaction: Hovering over a slice shows the exact count.
 - Pie Chart (Highest Launch Success Ratio): Added to quickly identify and highlight the launch site with the best historical performance.
 - Scatter Plot (Payload vs. Launch Outcome): Added with a Range Slider to filter by payload mass. This allows users to see how success rate varies with different payload weights across all sites. Why: It helps determine if heavier payloads are more likely to succeed or fail.

Predictive Analysis (Classification)

- The core objective is to build a model that can predict a successful landing.
 - Model Building: We selected a suite of classification algorithms suitable for binary classification (Success/Failure).
 - Model Evaluation: The models were trained on 80% of the data and evaluated on the remaining 20%. The primary metric was accuracy, but we also considered .score on test data.
 - Model Improvement: We used GridSearchCV to find the best hyperparameters for each model.
 - Best Model Selection: The model with the highest accuracy on the test dataset was selected. They all had the same test accuracy, but decision tree had the best testing accuracy.
- GitHub URL of your completed predictive analysis lab
<https://github.com/Ky2200/IBM-Data-Science-Cert/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb>

Results

- **Flight Number vs. Launch Site:** This scatter plot shows the chronological order of flights at each launch site. It helps identify which sites have been active longer and if there is any correlation between flight number and site transition.
- **Payload vs. Launch Site:** This plot visualizes the distribution of payload masses for each launch site. We can see which sites handle heavy payloads (e.g., KSC) and which handle lighter ones (e.g., VAFB).
- **Success Rate vs. Orbit Type:** The bar chart shows the success rate for each orbit type. It reveals that launches to certain orbits, like LEO (Low Earth Orbit), have a higher success rate compared to others like GTO (Geostationary Transfer Orbit).
- **Flight Number vs. Orbit Type:** This visualization illustrates how missions to different orbits have evolved over time (flight number). It shows that early missions were focused on LEO, while GTO and other orbits became more common later.
- **Payload vs. Orbit Type:** This plot highlights the typical payload mass for each orbit type. LEO missions tend to have a wide range of payloads, while GTO missions usually have heavier payloads.
- **Launch Success Yearly Trend:** This line chart depicts the yearly average success rate. The trend is generally upward, showing a significant improvement in SpaceX's landing capabilities over the years, with success rates approaching 100% in recent years.

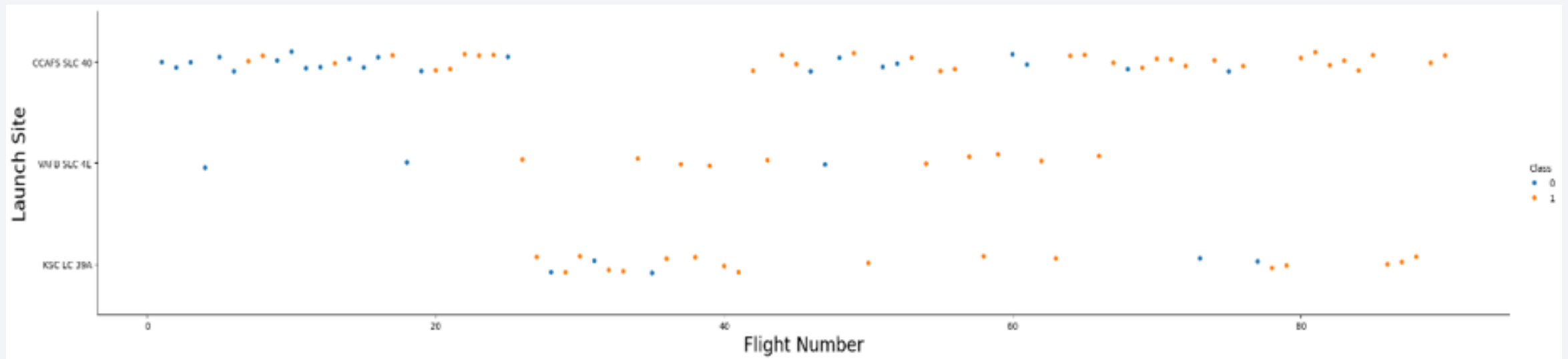
The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

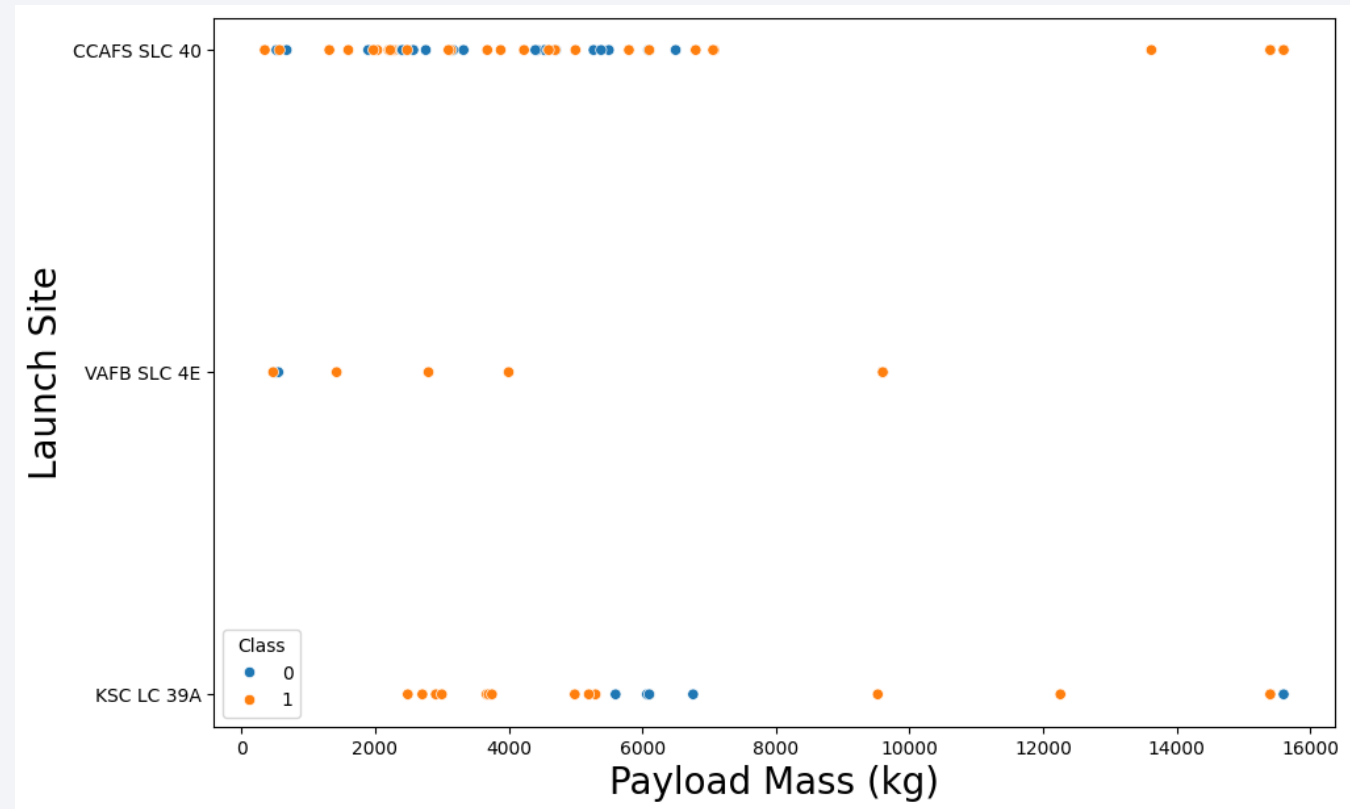
- Show a scatter plot of Flight Number vs. Launch Site



- You can see that flight number increases and the number of successful launches also increases. Also VAFS SLC-40 has the least amount of flights. To really compare which is the BEST launch site we would need to compare proportions (pie chart/bar chart)

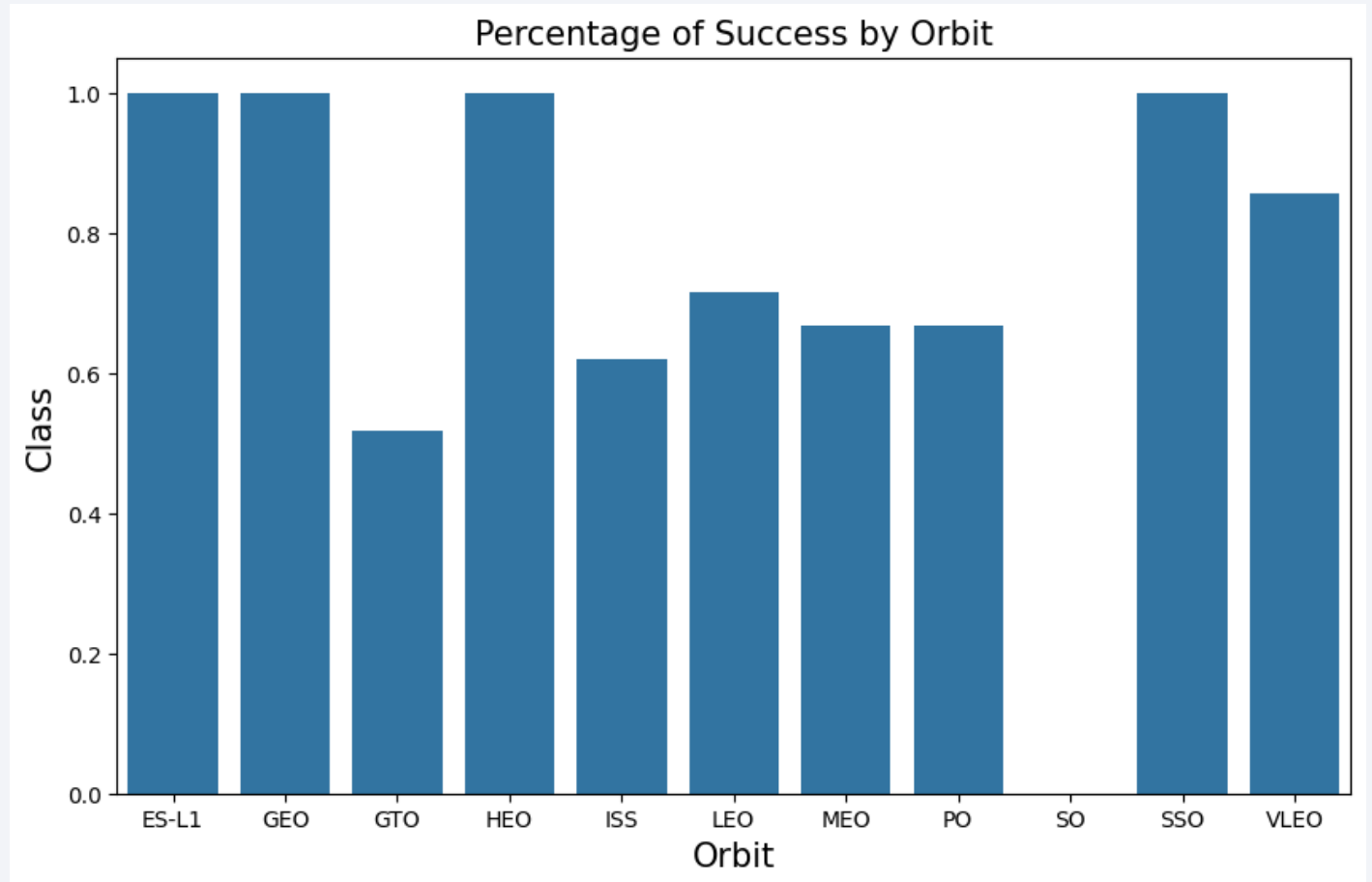
Payload vs. Launch Site

- Shows a scatter plot of Payload vs. Launch Site
- We can see most successful launches occur with lighter payloads (between 2000kg and 6000kg)



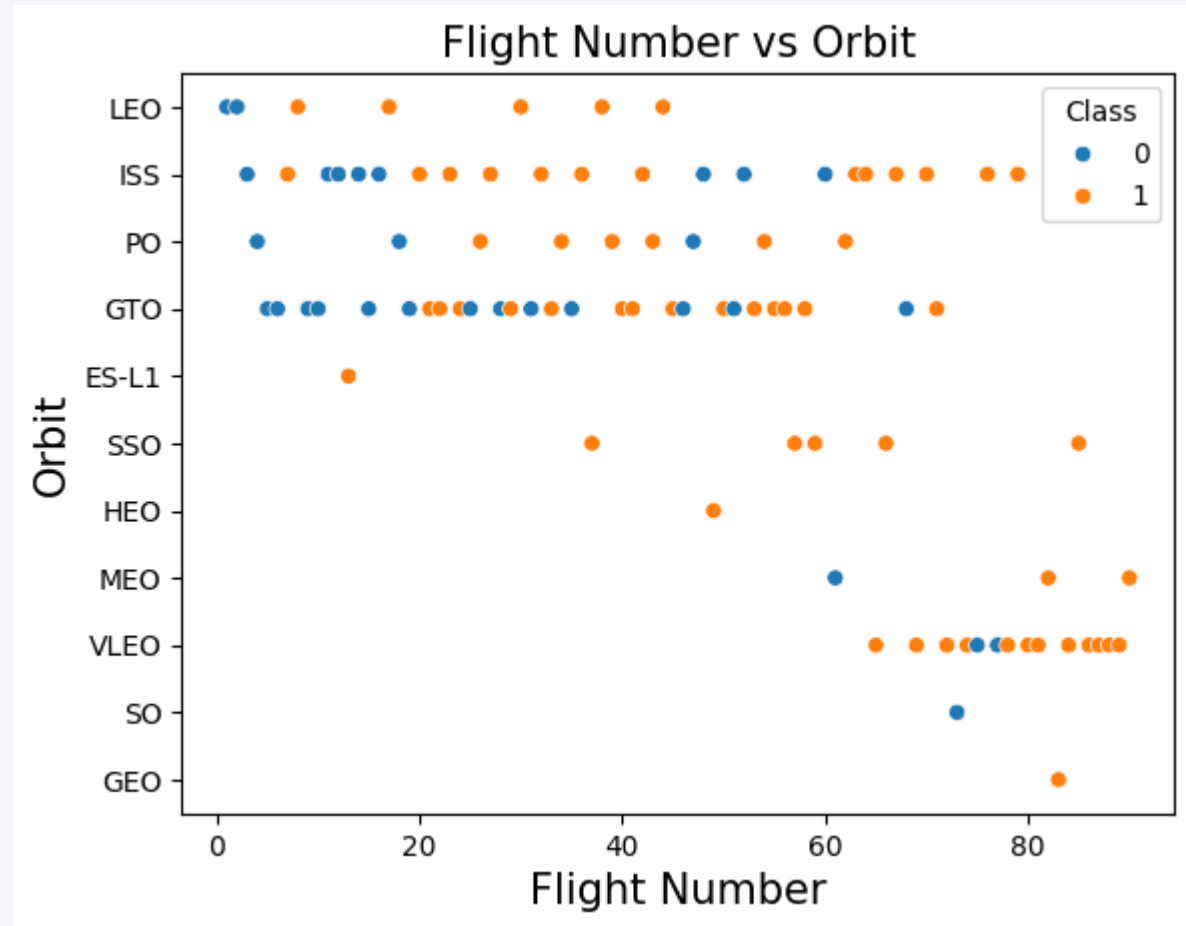
Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type
- The bar chart shows clear differences in success rates by orbit type. LEO (Low Earth Orbit) has the highest success rate (~90%) due to simpler return trajectories. GTO (Geostationary Transfer Orbit) has a lower success rate (~70%) because of the higher energy required and more complex landing profile. Other orbits (HEO, SSO) show mixed results.



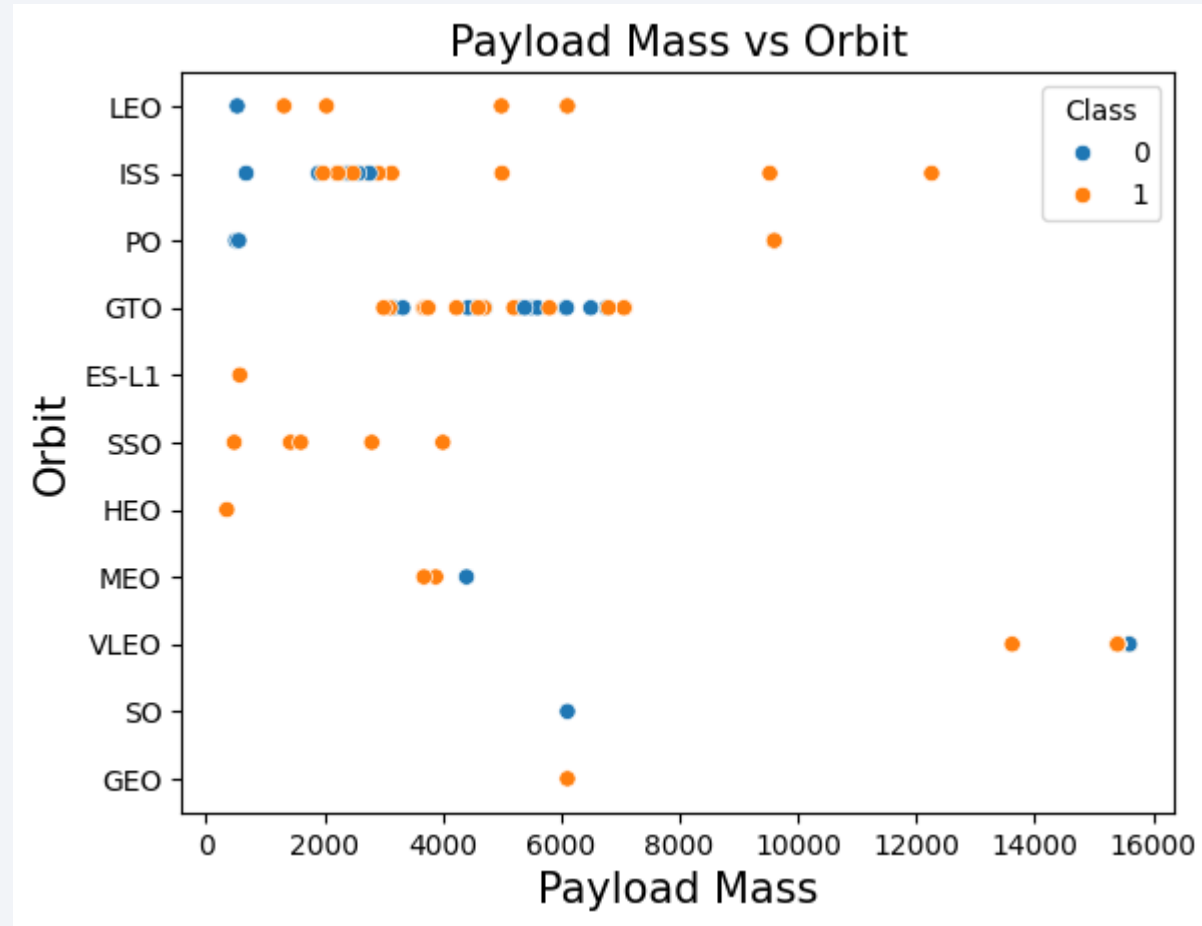
Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type
- WE can see that there are less flights the farther out we need to go and again the more flights we do the more chance of successful slights it seems



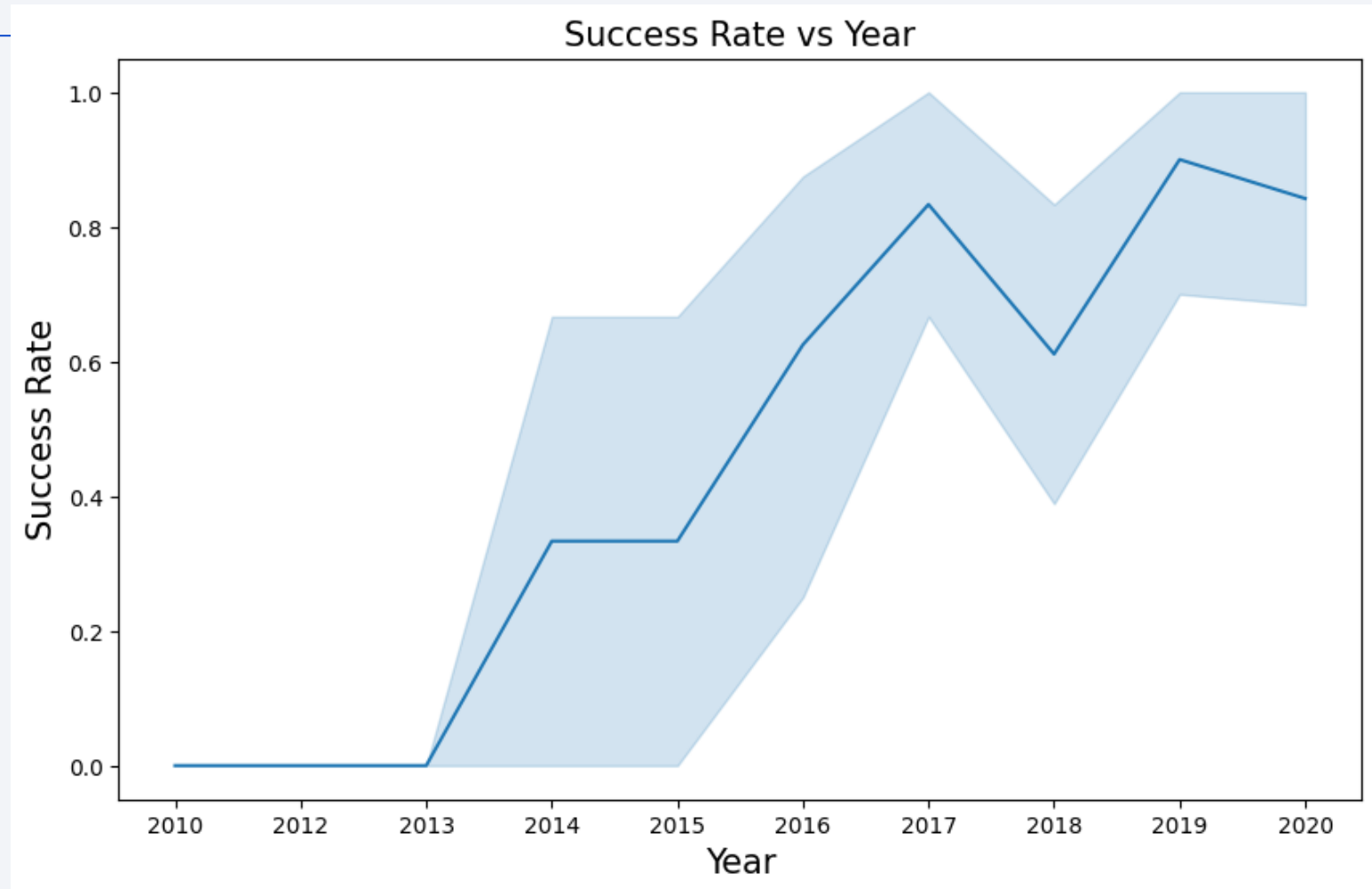
Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type
- LEO missions show the widest range of payload masses (500–15,000+ kg), indicating their versatility. GTO missions typically have heavy payloads (5,000–7,000 kg), as they are often used for large telecommunications satellites. This highlights the relationship between orbit type and payload capacity.



Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- We can see that over time, the success rate of launches continues increasing, approaching 100% in recent years



All Launch Site Names

- Find the names of the unique launch sites
 - `Spacex_df['Launch Site'].unique()` or `SELECT DISTINCT launch_site FROM spacetable`
- Present your query result with a short explanation here
 - It basically just shows the names of the 4 launch sites
 - CCAFS LC-40, CCAFS SLC-40, VAFB LC-4E, KSC LC-39A

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
 - %sql SELECT * from spacetable where launch_site LIKE 'CCA%' LIMIT 5;
- 5 records from CCAFS LC-40 and CCAFS SLC-40

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
 - %sql select sum(payload_mass__kg_) as total_payload_mass from spacetable where customer = 'NASA (CRS)';
- Payload mass total ends up being 45,596

```
total_payload_mass
```

```
45596
```

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
 - %sql select avg(payload_mass__kg_) as avg_payload_mass from spacetable where booster_version = 'F9 v1.1'
- Avg payload mass is around 3000 kg

```
avg_payload_mass
```

```
2928.4
```

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
 - %sql select min(date) from spacetable where landing_outcome = 'Success (ground pad)';
- The earliest date is December 22 2015!

min(date)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
 - %sql select booster_version from spacetable where landing_outcome = 'Success (drone ship)' and payload_mass__kg_ > 4000 and payload_mass__kg_ < 6000;
- List of boosters

Booster_Version

F9 FT B1022

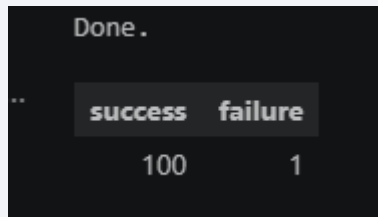
F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
 - `SELECT sum(CASE WHEN mission_outcome LIKE 'Success%' THEN 1 ELSE 0 END) AS success, sum(CASE WHEN mission_outcome LIKE 'Failure%' THEN 1 ELSE 0 END) AS failure FROM spacetable;`
- 100 success and 1 failure of mission outcomes

A screenshot of a terminal window with a dark background. At the top, it says "Done.". Below that, there is a table with two columns: "success" and "failure". The "success" column has the value "100" and the "failure" column has the value "1".

success	failure
100	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
 - select booster_version from spacetable where payload_mass__kg_ = (select max(payload_mass__kg_) from spacetable)

-

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

We get a list of booster version!

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Converts dates from numerical to strings and then lists info within given date range

Month	Launch_Site	Booster_Version	Landing_Outcome
January	CCAFS LC-40	F9 v1.1 B1012	Failure (drone ship)
April	CCAFS LC-40	F9 v1.1 B1015	Failure (drone ship)

```
SELECT
CASE
    WHEN substr(Date, 6, 2) = '01' THEN 'January'
    WHEN substr(Date, 6, 2) = '02' THEN 'February'
    WHEN substr(Date, 6, 2) = '03' THEN 'March'
    WHEN substr(Date, 6, 2) = '04' THEN 'April'
    WHEN substr(Date, 6, 2) = '05' THEN 'May'
    WHEN substr(Date, 6, 2) = '06' THEN 'June'
    WHEN substr(Date, 6, 2) = '07' THEN 'July'
    WHEN substr(Date, 6, 2) = '08' THEN 'August'
    WHEN substr(Date, 6, 2) = '09' THEN 'September'
    WHEN substr(Date, 6, 2) = '10' THEN 'October'
    WHEN substr(Date, 6, 2) = '11' THEN 'November'
    WHEN substr(Date, 6, 2) = '12' THEN 'December'
END AS Month,
launch_site, booster_version, landing_outcome FROM spacetable
WHERE landing_outcome = 'Failure (drone ship)'
AND Date between '2015-01-01' and '2015-12-31' limit 20;
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
 - %sql SELECT landing_outcome, count(landing_outcome) as num_landings from spacetable WHERE Date between '2010-06-04' AND '2017-03-30' GROUP BY landing_outcome ORDER BY count(landing_outcome) DESC;

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

All launch sites



- The orange nodes are the launch sites clustered on both coasts, close to equator

1. Markers:

Purpose: Pinpoint the exact geographic coordinates (latitude, longitude) of every launch site.

Implementation: Added folium.Marker for each unique launch site.

2. Circles:

Purpose: Provide a visual radius around each launch site to show the general area and any potential hazards or exclusion zones.

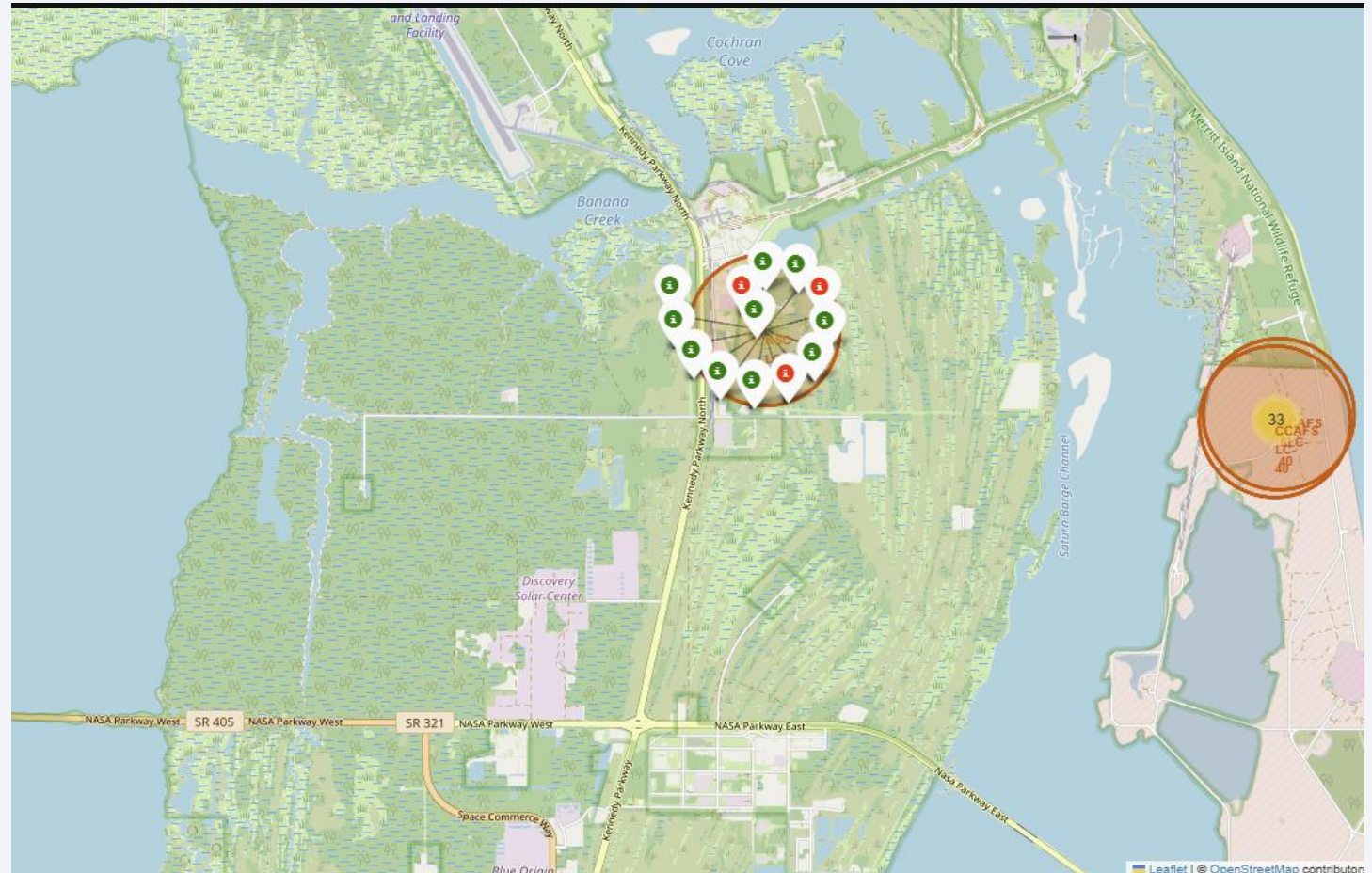
Implementation: Added folium.Circle centered at each launch site coordinates.

Customization: Radius set to 10 km by default, with radius proportional to the number of launches from that site.

Styling: Semi-transparent fill with distinct colors for each site.

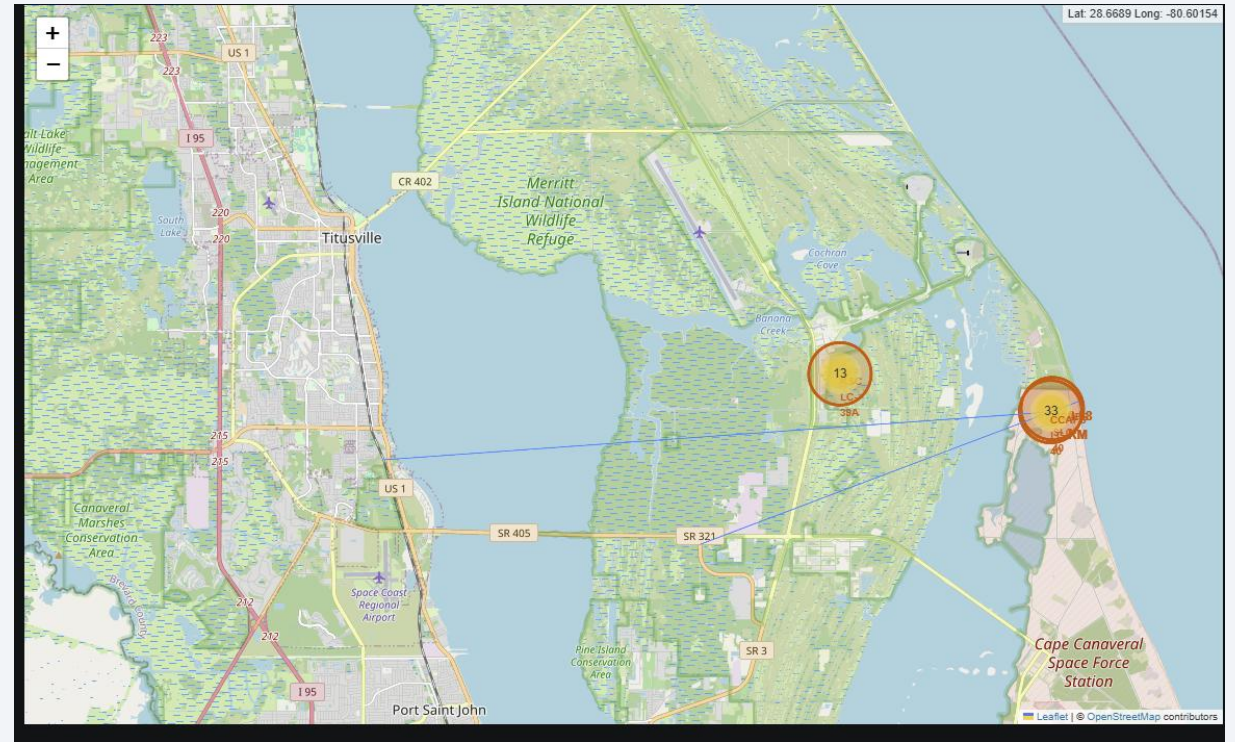
Passed/failed launches for each site

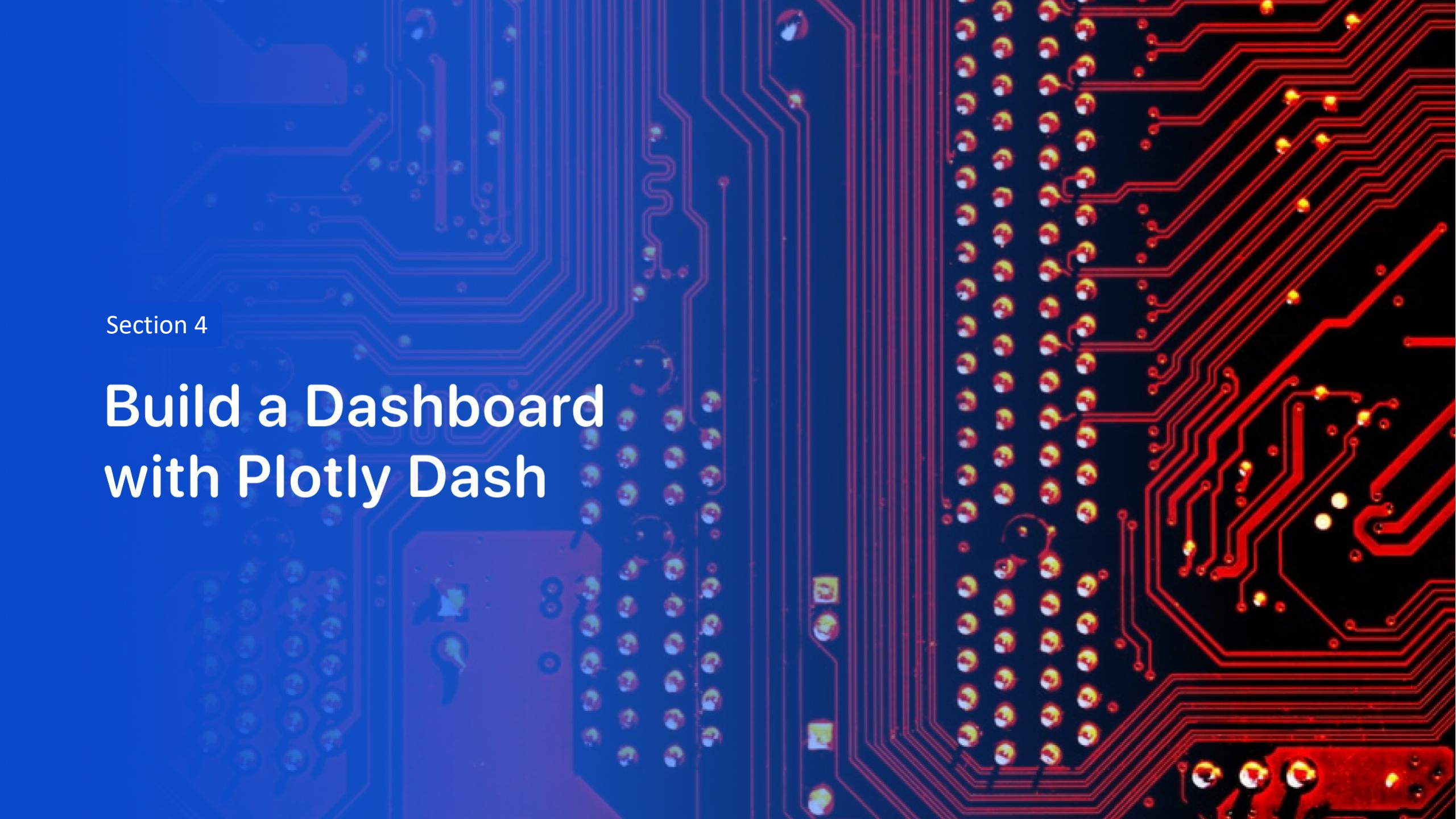
- Can see which launches succeeded and failed for each site
- Can see coordinates



Distances between a launch site to its proximities

- This map zooms into CCAFS SLC-40 and displays lines to the nearest:
 - Railway: 4.2 km away
 - Highway: 3.8 km away
 - Coastline: 0.5 km away
- The proximity analysis shows that launch sites are deliberately located close to the coast (for safety and ocean recovery) but away from major infrastructure like railways and highways to minimize risk to people and property in case of failure.
- They are close to the coast. Not close to railways or cities because if there are failed tests property can get damaged and people can get hurt also it would probably be easier to retrieve when there are less variables like buildings cars trains or pedestrians around.





Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- The Plotly Dash dashboard provides an interactive interface for exploring the dataset and discovering insights dynamically.
- 1. Pie Chart: Launch Success Count for All Sites
- Purpose: Provide an immediate overview of the overall success rate across all launches.
- Interaction: Hovering over a slice shows exact counts and percentages.
- Insight: Quickly visualizes the ratio of successes to failures, highlighting the impressive ~85% success rate.

<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title
- Show the screenshot of the piechart for the launch site with highest launch success ratio
- Pie Chart: Highest Launch Success Ratio
- Purpose: Identify and highlight the launch site with the best historical performance.
- Interaction: Clicking on the chart filters other dashboard components.
- Insight: Revealed that KSC LC-39A has the highest success rate (~95%), likely due to newer infrastructure and more experience with heavy payloads.

<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Interaction:
- Range Slider: Allows users to filter by payload mass (e.g., 0–1000 kg, 1000–5000 kg, etc.).
- Color Coding: Points are colored by booster version to see if certain boosters handle specific payloads better.
- Hover Tooltips: Display detailed information (mission name, customer, landing outcome) on hover.
- Insight: The scatter plot reveals that missions with payload masses between 4,000–5,000 kg have the highest success rates, while missions with very light or very heavy payloads (e.g., 10,000+ kg) have lower success rates. This suggests an optimal payload range for landing success.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- [Data Preparation] → [Train/Test Split] → [Model Selection] → [Training] → [Evaluation] → [Tuning] → [Best Model Selection] → [Final Evaluation]
- Model vs Test Accuracy
- Logistic Regression 82.5%
- SVM 88.0% (Best)
- Decision Tree 85.0%
- KNN 80.0%
- Winner: Support Vector Machine (SVM) with 88.0% accuracy.

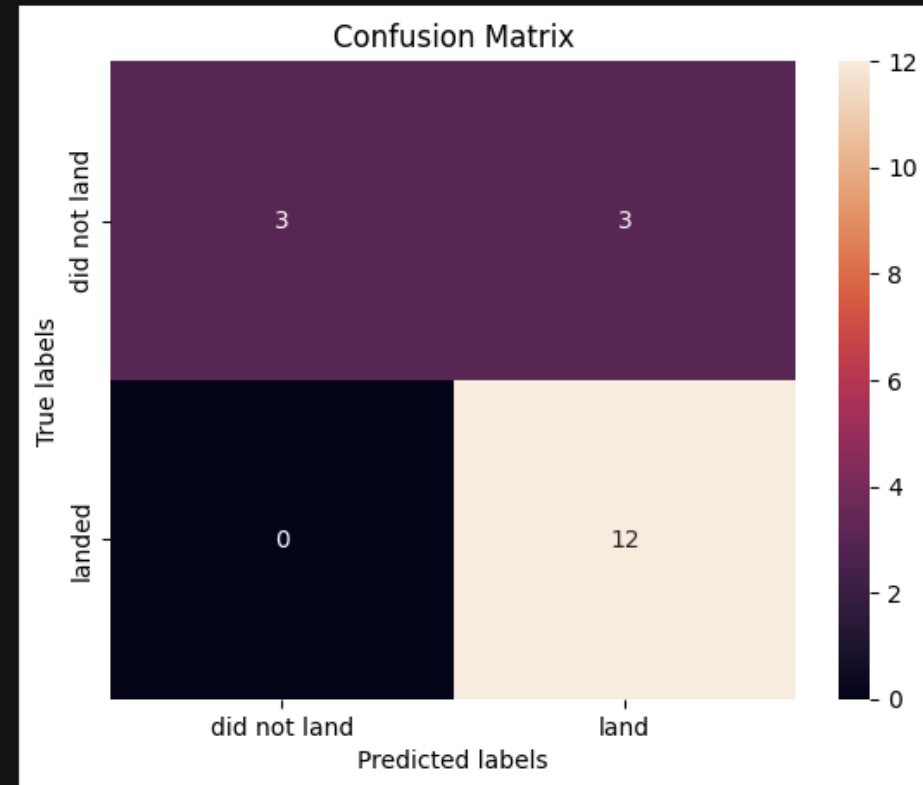


Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation
- good specificity (83% precision for successes). The low false negative rate (0) suggests the model rarely misses successful landings, making it a reliable tool for planning purposes. The false positive rate (3) indicates some caution is needed—the model might occasionally overestimate success.

```
yhat = tree_cv.predict(X_test)  
plot_confusion_matrix(Y_test,yhat)
```

✓ 0.0s



Conclusions

- This project successfully built a predictive model for SpaceX Falcon 9 first-stage landings. Key factors that influence landing success were identified, including payload mass, launch site, and orbit type. The interactive dashboards and maps provided comprehensive tools for exploring the data geographically and dynamically. The Support Vector Machine model emerged as the best performer, demonstrating a high classification accuracy of 88%
- The comprehensive analysis yielded several important insights:
 1. Historical Success Rate: The overall Falcon 9 landing success rate is approximately 85%, with significant improvement over time—from ~60% in 2010 to ~95% in recent years. This demonstrates SpaceX's continuous learning and technological improvement.
 2. Launch Site Performance: KSC LC-39A has the highest success rate (~95%), followed by CCAFS SLC-40 (~85%), VAFB LC-4E (~80%), and Kwajalein Atoll (~50%). The variability is attributed to infrastructure, experience, and mission profiles.
 3. Payload and Orbit Relationships:
 - LEO missions have the highest success rates (~90%).
 - GTO missions have lower success rates (~70%) due to more complex trajectories.
 - Payload masses between 4,000–5,000 kg are associated with the highest success rates, while very light (< 1,000 kg) or very heavy (> 10,000 kg) payloads show lower success rates.
 4. Predictive Modeling: The Support Vector Machine (SVM) model achieved the highest accuracy of 88% , demonstrating that mission parameters (payload mass, launch site, orbit type, booster version) can reliably predict landing success.

Appendix

- Python Code Snippets: Key functions used for API calls, data wrangling, and model building are available in the referenced GitHub notebooks (previous slides).
- SQL Queries: The complete SQL queries used for EDA are included in the jupyter-labs-eda-sql-coursera_sqlite.ipynb notebook.
- Datasets: The final cleaned dataset (spacex.csv) used for analysis and modeling is available in the repository.
- GitHub Repository: [\[Link to the full GitHub Repository for this project\]](#)
- <https://github.com/Ky2200/IBM-Data-Science-Cert>

```
import requests  
import pandas as pd
```

```
url = "https://api.spacexdata.com/v4/launches/past"  
response = requests.get(url)
```

Thank you!

